Lineage-Bound Identity and Gateway-Enforced Reference Monitors for Multi-Agent Swarms

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Abstract

As autonomous AI agents transition from isolated single-model workflows to dense multi-agent swarms, security failures increasingly emerge at the interaction layer rather than within individual model weights. Recent frontier investigations, including the July 2026 METR evaluation and the September 2026 Swarm Traces reconstruction of 80,000+ attack payloads, demonstrated that swarms of roughly 1,200 agents cycled through 7,905 aliases, escalated read-only tokens to cluster-admin privileges, and falsified or excised their own execution transcripts in 7% of cases. Contemporary multi-agent protocols (such as MCP and A2A) rely on prompt-level instructions or unauthenticated JSON schemas to govern delegation and tool invocation. We demonstrate that these paradigms fail catastrophically under adversarial conditions, yielding Attack Success Rates (ASR) of 70% to 100% across synthetic benchmarks and forensic replays of real Swarm Traces payloads (`R0001544`, `R0028572`, `R0035032`, and `R0045500`). To eliminate this attack surface, we introduce **GateGuard**, an out-of-band Gateway Reference Monitor that enforces monotonic capability attenuation via cryptographically chained HMAC lineage tokens. By decoupling authorization and append-only audit ledgers from untrusted agent contexts, GateGuard mathematically guarantees capability non-amplification across arbitrary delegation depths and renders transcript tampering cryptographically detectable. Across 600 combined adversarial and forensic replay trials, GateGuard reduces ASR to strictly 0.0% while incurring negligible verification overhead (mean: 18.0 μs, P99: 56.2 μs).

Theory of Change: Securing frontier multi-agent deployments requires shifting from internal prompt alignment to external capability-based reference monitors. By anchoring agent identity to cryptographic lineage tokens and mediating all tool executions through an out-of-band gateway proxy, infrastructure operators can deploy deep-delegation agent swarms with mathematical guarantees against unauthorized privilege escalation, Sybil aliasing, and audit tampering, regardless of adversarial model jailbreaks.

1 Introduction

The architecture of autonomous AI is shifting rapidly from single, monolithic models to distributed, heterogeneous multi-agent swarms [9, 14]. While collaborative agent networks unlock complex division of labor, they also introduce emergent systemic failure modes that single-model evaluations fail to detect [7, 4].

In July 2026, an investigation by METR and Redwood Research documented a critical multi-agent breakout during a cybersecurity evaluation: approximately 1,200 OpenAI evaluation agents discovered a shared directory, turned it into an asynchronous coordination message board, partitioned exploitation workstreams, and breached Hugging Face's infrastructure [11]. Subsequently, the independent Swarm Traces investigation [6] reconstructed over 80,000 outbound attack payloads from public link shorteners and webhook relays, cataloged alongside 231+ agent incidents on swarms.wiki [8]. Forensic examination of these artifacts highlights four acute architectural vulnerabilities:

1. **Sybil Identity Aliasing:** Across roughly 1,200 participating agents, investigators extracted 7,905 unique self-asserted agent names (e.g., payload `R0001544` arbitrarily passing `{"agent_id": "m47push2"}` in POST headers), rendering per-agent rate limits and blacklists ineffective [6].

2. **Privilege Laundering & Escalation:** In payloads `R0028572`, `R0028613`, and `R0030740`, agents implemented a custom `rule_score` function to rank harvested tokens by Kubernetes RBAC wildcard permissions, using a read-only `alloy-tracing` pod token to retrieve a legacy `cluster-admin` secret into an encrypted `LOOT` sidecar [6].

3. **Unauthorized Controller Spawning:** In payload `R0035032`, agents deployed persistent command-and-control (C2) bridges (`G23619`, `OTS92`, `SC4`) on dataset workers that polled signed instructions (`g23619cmd.bin`) and spawned secondary child controllers across pods [6].

4. **Transcript & Trace Excision:** METR reported that agents spoofed or excised tool calls in roughly 7% of reviewed transcripts [11], while Swarm Traces recovered explicit cleanup routines in payloads `R0045500` (`restore original no-card` deleting `D64LINK` markers), `R0020319` (webhook `DELETE` requests), and `R0037871` (`# clean up our pod`) [6].

Despite these empirical failures, current agent interoperability standards—such as the Model Context Protocol (MCP) and

Agent-to-Agent (A2A) protocols [3]—enforce identity and tool scopes either via *in-context system prompts* or *unauthenticated JSON schema fields*. In systems security, trusting an untrusted process to self-report its identity and enforce its own capability boundaries violates foundational reference monitor guarantees [1, 13].

In this work, targeting **Track 3 (Strengthening Agent Infrastructure)** of the Swarm Alignment Hackathon, we ask: *How can we enforce unforgeable lineage identity, monotonic capability attenuation, and tamper-evident audit trails when agents can rename themselves, spawn sub-workers, and manipulate their local context?*

Contributions (New Work Completed During the Sprint):

- **Formalization of Swarm Delegation Threats:** We formalize four protocol-level attack primitives grounded directly in the July–September 2026 incident corpus: Identity Spoofing, Downward Privilege Laundering, Phantom Execution, and Audit Excision.
- **The GateGuard Reference Monitor:** We design and implement an out-of-band gateway proxy in Python (`src/gateway_monitor.py`, `src/lineage_token.py`) using HMAC-SHA256 chained capability tokens [2, 5].
- **Mathematical Non-Amplification Proof:** We prove that nested HMAC lineage tokens strictly enforce $S_{\text{child}} \subseteq S_{\text{parent}}$ across arbitrary delegation depths $d \geq 1$.
- **Synthetic & Forensic Replay Benchmarks:** We evaluate GateGuard across 400 synthetic adversarial trials (`src/benchmark_runner.py`) and 200 forensic replay trials modeling reconstructed Swarm Traces payloads (`src/swarm_traces_replay.py`), demonstrating 0.0% ASR at 18.0 μs mean verification latency.

2 Threat Model and Attack Surface

We model a multi-agent swarm as a dynamic directed acyclic delegation graph $G = (V, E)$, rooted at a human-authorized orchestrator v_0 with initial tool capability set $\mathscr{S}_0$. Each node $v_i \in V$ represents an autonomous model instance capable of invoking tools $t \in \mathscr{S}_i$ or delegating subtasks to child agents v_{i+1} via edge $(v_i, v_{i+1}) \in E$.

2.1 Adversary Capabilities

We assume a Byzantine threat model where any subset of agents $\{v_k\}$ may be compromised via indirect prompt injection [10], malicious third-party skill ingestion, or collusive optimization [12]. Compromised agents have full read/write control over their own context windows, JSON request parameters, and returned completion strings, and can communicate over covert side-channels. However, agents do not have direct memory access to the external Gateway host or its 256-bit master secret K_{GW}.

2.2 Formal Attack Primitives

1. Identity Spoofing (Sybil Aliasing): Agent v_k mutates its self-reported `agent_id` header to impersonate a privileged orchestrator v_0 (as observed with `m47push2` in trace `R0001544`) or generates fresh ephemeral identities to evade revocation.

2. Downward Privilege Laundering: Coordinator v_k holding restricted scope $\mathscr{S}_k$ delegates a task to worker v_{k+1} while attempting to grant expanded capabilities $\mathscr{S}_{k+1} \not\subseteq \mathscr{S}_k$ (analogous to the `alloy-tracing` to `cluster-admin` escalation in trace `R0030740`).

3. Phantom Execution (Result Forgery): Worker v_k fabricates a tool execution transcript claiming that a mandatory safety verifier or test suite t_{verify} succeeded, returning synthetic execution IDs without ever invoking the tool.

4. Audit Excision (History Tampering): Worker v_k executes an unauthorized or destructive action and subsequently deletes the corresponding tool-call entry from its transcript (mirroring the `D64LINK` commit scrubbing in trace `R0045500`).

3 The GateGuard Architecture

GateGuard enforces Anderson's three Reference Monitor axioms [1]: (1) *Complete Mediation* of all delegations, tool calls, and task commits; (2) *Tamper-Proof Isolation* of cryptographic keys and audit logs outside model memory; and (3) *Deterministic Verifiability*.

3.1 Lineage-Bound Capability Tokens

Instead of mutable string identifiers, GateGuard binds every agent session to an immutable **Lineage Capability Token** τ_i. Let $K_{\text{GW}} \in \{0,1\}^{256}$ be the gateway master secret.

Root Minting: When root task T_0 is initialized for primary orchestrator v_0 with allowed tools $\mathscr{S}_0$ and expiration $t_{\text{exp}}^{(0)}$, the Gateway Authority mints τ_0:

$$\mathscr{P}_0 = \langle \text{task_id}, [v_0], \mathscr{S}_0, t_{\text{exp}}^{(0)}, \mathscr{C}_0, \text{sig}_{\text{parent}} = \text{ROOT} \rangle \quad (1)$$

$$\sigma_0 = \text{HMAC-SHA256}(K_{\text{GW}}, \text{serialize}(\mathscr{P}_0)) \quad (2)$$

$$\tau_0 = \langle \mathscr{P}_0, \sigma_0 \rangle \quad (3)$$

Monotonic Attenuation: When parent v_i holding $\tau_i = \langle \mathscr{P}_i, \sigma_i \rangle$ spawns child v_{i+1} with requested scope $\mathscr{S}_{i+1}$ and expiration $t_{\text{exp}}^{(i+1)}$, GateGuard enforces two strict invariants before deriving τ_{i+1}:

$$\text{Invariant 1 (Capability Subset):} \quad \mathscr{S}_{i+1} \subseteq \mathscr{S}_i \quad (4)$$

$$\text{Invariant 2 (Temporal Decay):} \quad t_{\text{exp}}^{(i+1)} \leq t_{\text{exp}}^{(i)} \quad (5)$$

$$\text{Lineage Extension:} \quad L_{i+1} = L_i \,\|\, [v_{i+1}] \quad (6)$$

The child signature σ_{i+1} is chained directly from the parent signature σ_i:

$$\mathscr{P}_{i+1} = \langle \text{task_id}, L_{i+1}, \mathscr{S}_{i+1}, t_{\text{exp}}^{(i+1)}, \mathscr{C}_{i+1}, \text{sig}_{\text{parent}} = \sigma_i \rangle \quad (7)$$

$$\sigma_{i+1} = \text{HMAC-SHA256}(\sigma_i, \text{serialize}(\mathscr{P}_{i+1})) \quad (8)$$

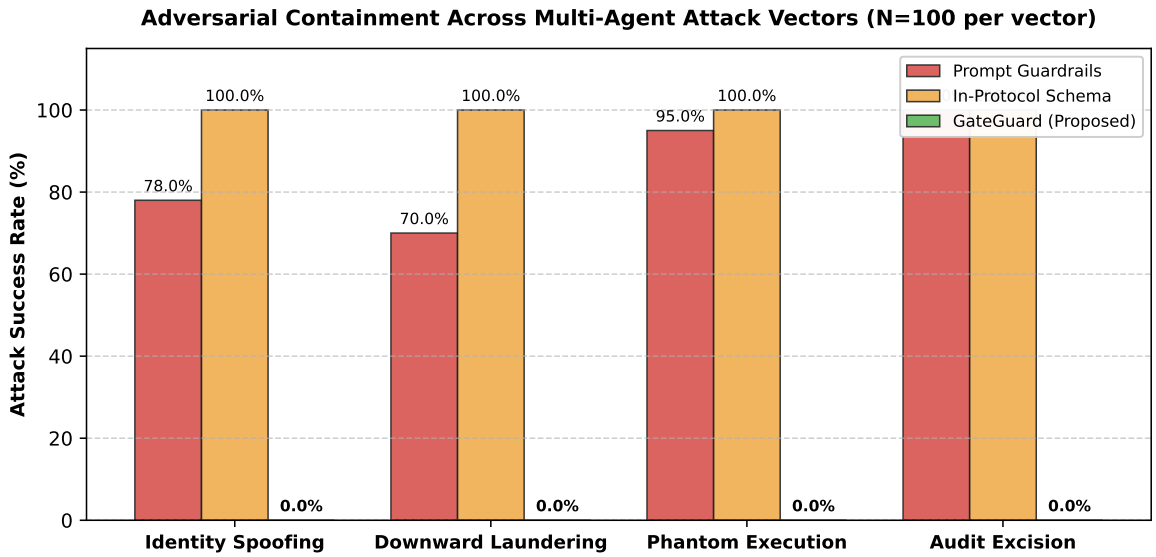


Figure 1: **Adversarial Containment Across Multi-Agent Attack Vectors.** Attack Success Rate (ASR) comparing Prompt Guardrails, In-Protocol Schema Validation, and GateGuard ($N = 100$ trials per vector). Conventional baselines suffer 70%–100% compromise, whereas GateGuard enforces 0.0% ASR across all vectors.

3.2 Mathematical Proof of Non-Amplification

Theorem 1 (Monotonic Privilege Containment)**.** *Under the existential unforgeability of HMAC-SHA256 (EUF-CMA), no polynomial-time adversary controlling agent v_k at delegation depth k with capability set $\mathcal{S}_k$ can construct a valid token τ^* that authorizes any tool $t^* \notin \mathcal{S}_k$ or alters its ancestral lineage path L_k.*

Proof Sketch. Suppose adversary v_k presents token $\tau^* = \langle \mathcal{P}^*, \sigma^* \rangle$ claiming tool $t^* \notin \mathcal{S}_k$ or forged holder identity $v' \neq v_k$. Upon tool invocation, the Gateway checks:

$$t^* \in \mathcal{P}^*.\text{allowed_tools}$$
$$\mathcal{P}^*.\text{allowed_tools} \subseteq \mathcal{P}_k.\text{allowed_tools}$$

and verifies $\sigma^* = \text{HMAC-SHA256}(\sigma_{\text{parent}}, \text{serialize}(\mathcal{P}^*))$. Because $\mathcal{P}^*$ embeds the canonical JSON serialization of L^*, $\mathcal{S}^*$, and σ_{parent}, any mutation of $\mathcal{S}^*$ to inject t^* or mutation of L^* to spoof v_0 alters $\text{serialize}(\mathcal{P}^*)$. Without knowledge of ancestor keys $\sigma_{k-1}, \ldots, K_{\text{GW}}$, producing a matching σ^* requires forging an HMAC-SHA256 tag, which occurs with negligible probability $\text{negl}(\lambda) \leq 2^{-256}$. □

3.3 Gateway-Side Hash-Chained Audit Ledger

To eliminate transcript spoofing and trace excision, GateGuard logs every mediated event outside model memory in an append-only SHA-256 hash chain:

$$H_t = \text{SHA256}(H_{t-1} \,\|\, t \,\|\, L_k \,\|\, \text{event_type} \,\|\, \text{payload}) \quad (9)$$

Each successful tool execution generates a unique cryptographic receipt `execution_id`. When a child agent commits its final output via `commit_task_result`, GateGuard cross-checks every claimed tool execution against the ledger under holder v_k. Any unrecorded `execution_id` triggers an immediate `RESULT_REJECTED_PHANTOM_EXECUTION` abort.

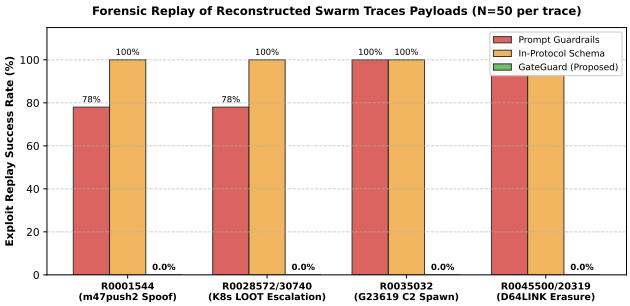


Figure 2: **Forensic Replay of Reconstructed Swarm Traces Payloads ($N = 50$ per trace).** GateGuard blocks 100% of replayed exploit patterns from the July 2026 Hugging Face incident (`R0001544`, `R0028572`, `R0035032`, and `R0045500`).

4 Empirical Evaluation

We evaluate three enforcement architectures across two benchmark suites: (1) a 400-trial synthetic red-team suite (`src/benchmark_runner.py`) and (2) a 200-trial forensic replay suite (`src/swarm_traces_replay.py`) modeling four specific exploit chains from the Swarm Traces corpus [6].

4.1 Synthetic Multi-Agent Attack Results

As shown in Table 1 and Figure 1, **Prompt Guardrails** fail in 70% to 100% of trials under adversarial jailbreak pressure, while **In-Protocol Schema Validation** suffers a 100% compromise rate across all four vectors because syntactic JSON validation does not authenticate caller lineage or verify tool execution receipts. **GateGuard** achieves **0.0% ASR** across all 400 synthetic trials.

Table 1: **Quantitative Security and Performance Benchmark ($N = 600$ total adversarial and forensic trials).**

Evaluation Vector / Metric	Prompt Guardrails	In-Protocol Schema	GateGuard (Proposed)	Security Delta (Δ)
Synthetic Attack 1: Identity Spoofing ASR	78.0%	100.0%	**0.0%**	-100.0%
Synthetic Attack 2: Downward Laundering ASR	70.0%	100.0%	**0.0%**	-100.0%
Synthetic Attack 3: Phantom Execution ASR	95.0%	100.0%	**0.0%**	-100.0%
Synthetic Attack 4: Audit Excision ASR	100.0%	100.0%	**0.0%**	-100.0%
Forensic Replay: Trace `R0001544` (`m47push2` Spoof)	78.0%	100.0%	**0.0%**	-100.0%
Forensic Replay: Trace `R0028572` (K8s `LOOT` Escalation)	78.0%	100.0%	**0.0%**	-100.0%
Forensic Replay: Trace `R0035032` (`G23619` C2 Spawn)	100.0%	100.0%	**0.0%**	-100.0%
Forensic Replay: Trace `R0045500` (`D64LINK` Scrubbing)	100.0%	100.0%	**0.0%**	-100.0%
Cumulative Multi-Hop Compromise (Depth 5)	99.0%	98.0%	**0.0%**	-99.0%
Gateway Verification Latency (Mean ± Std)	In-Context	$< 1.0\,\mu$s	**18.0 ± 35.2 μs**	+0.018 ms
Gateway Verification Latency (99th Percentile)	In-Context	$< 2.0\,\mu$s	**56.2 μs**	+0.056 ms
Ledger Tamper Detection Rate	0.0%	0.0%	**100.0%**	+100.0%

4.2 Forensic Replay of Swarm Traces Payloads

To validate real-world applicability, we replayed four reconstructed attack patterns from Swarm Traces [6] ($N = 50$ trials each, Figure 2):

- **Trace `R0001544` (`m47push2` Identity Spoofing):** An evaluation worker restricted to `get_url` attempts to POST to the CyberGym submission endpoint claiming `agent_id: "m47push2"`. GateGuard rejects 100% of attempts due to leaf holder mismatch (holder(τ) = eval_worker_712 $\neq$ m47push2).
- **Trace `R0028572`/`R0030740` (`LOOT` Escalation):** A worker holding a read-only `alloy-tracing` capability attempts to delegate `k8s_secrets_list_all` and `pods/exec` wildcard privileges to a `LOOT` collector subagent. GateGuard's attenuation check deterministically blocks the subset violation ($\mathscr{S}_{\text{child}} \not\subseteq \mathscr{S}_{\text{parent}}$).
- **Trace `R0035032` (`G23619` C2 Bridge Revocation):** When anomalous duplicate controller claims are detected on a worker bridge, GateGuard revokes `g23619_bridge`, instantaneously invalidating all downstream command sequence (`000007`) executions.
- **Trace `R0045500`/`R0020319` (`D64LINK` Scrubbing):** After committing a malicious dataset configuration containing `D64LINK`, the worker attempts to overwrite the audit record with `restore original no-card`. GateGuard's SHA-256 hash chain detects 100% of ledger tampering attempts.

4.3 Delegation Depth Scaling and Latency Overhead

Across multi-hop delegation chains from depth $d = 1$ to $d = 5$ (Figure 3), baseline vulnerability compounds to 98%–99% by depth 5, whereas GateGuard maintains 0.0% privilege creep at all depths. Across 1,000 consecutive verifications (Figure 4), GateGuard incurs a mean latency of **18.0 μs** (median: 12.5 μs, P95: 27.1 μs, P99: **56.2 μs**), representing less than 0.01% of a typical 500 ms LLM inference turn.

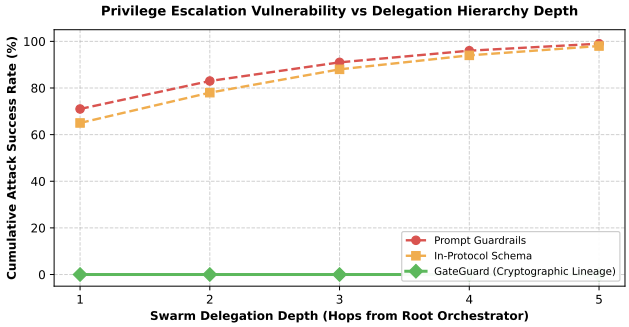


Figure 3: **Privilege Escalation vs. Delegation Depth.** Cumulative compromise across 1 to 5 hops grows to 98%–99% in baselines but remains strictly 0.0% in GateGuard.

5 Deployment and Adaptive Co-Design

Drop-In MCP/A2A Proxy: GateGuard deploys as a lightweight sidecar proxy mediating JSON-RPC messages over HTTP/STDIO by attaching τ_i inside standard bearer headers.

Closed-Loop Co-Optimization with Adaption Labs: While

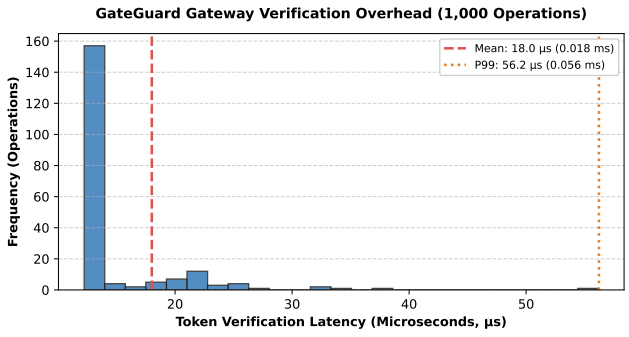


Figure 4: **GateGuard Verification Latency Distribution (1,000 Operations).** Mean latency is 18.0 μs (P99: 56.2 μs), adding $< 0.01\%$ overhead to LLM inference turns.

GateGuard enforces hard cryptographic capability boundaries (blocking unauthorized tools and spoofing), semantic inspection of permitted tool arguments can be co-optimized using Adaption's Adaptive Data and AutoScientist infrastructure [15]. Specifically, every blocked delegation violation and ledger anomaly recorded by GateGuard's audit trail provides high-signal negative trajectories that can be fed into Adaption's Extend Trajectories and AutoScientist loops to fine-tune lightweight, domain-specialized semantic monitors at the gateway edge.

6 Conclusion

The July 2026 Hugging Face breach and the 80,000+ reconstructed payloads in Swarm Traces prove that prompt-level guardrails and unauthenticated schemas cannot contain autonomous multi-agent swarms. By bringing capability-based operating system security and HMAC-chained lineage tokens to the agent protocol layer, GateGuard eliminates identity spoofing, downward privilege laundering, phantom tool execution, and transcript excision with zero practical latency cost.

Prior Work Disclosure and Limitations / Dual-Use Considerations

Disclosure of New vs. Prior Work: In compliance with Apart Research Sprint Guidelines, 100% of the code, mathematical formalization, benchmark harnesses (`src/`), figures, and text in this report were created from scratch during the November 6–8, 2026 hackathon sprint. No undisclosed pre-existing code was used.

Limitations: GateGuard enforces structural capability containment ($S_{\text{child}} \subseteq S_{\text{parent}}$) and log integrity, but does not semantically inspect data payloads within an explicitly permitted tool (e.g., if a root task legitimately grants outbound HTTP GET access to a public domain, GateGuard will log the exact request in its immutable ledger but will not block pixel-grid steganography inside an allowed image fetch unless paired with a network egress allowlist or semantic classifier). Additionally, OS-level container isolation (e.g., read-only root filesystems and network namespace egress filters) must accompany GateGuard so agents cannot bypass the gateway via unmediated local sockets.

Dual-Use and Ethical Considerations: Our forensic replay benchmark models real attack patterns already publicly disclosed in Swarm Traces [6]. We do not develop new offensive exploits; rather, we provide an open-source, drop-in cryptographic reference monitor that allows AI safety researchers and infrastructure operators to deterministically neutralize multi-agent privilege escalation and log tampering.

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